Laboratory records. Verifiable reports.

Block-Chain Powered Laboratory Records and Verification System

RHU of M’lang, North Cotabato

01

PROCESS

From request to released PDF

02

BLOCKCHAIN

From fingerprint to evidence

03

EVALUATION

Technical observations + survey

The system adds evidence to a laboratory report

Manage the workflow

Link patients, orders, specimens, results and report approvals in one process.

Preserve the artifact

Keep an official released PDF and the exact fingerprint associated with its version.

Verify later

Check the report’s status, compare its PDF and inspect committed blockchain evidence.

Core question: Does this document match the issued report and its recorded history?

Paper: pp. 3–5, 18–21.

Three access paths support different tasks

Authorized personnel

Register → process → review

Sign → approve → release

Actions depend on permissions.

Patients

Sign in to the patient portal.

View eligible released reports associated with their account.

Public verifier

Scan a QR or enter a code.

See limited verification status; compare a submitted PDF.

Patient data and official PDF access remain protected by the application.

From laboratory request to released report

01
Register patient
Identify the patient

→

02
Create order
Select examinations

→

03
Handle specimen
Register and track sample

→

04
Enter results
Record laboratory findings

05
Verify results
Review before reporting

→

06
Prepare report
Create result snapshot

→

07
Sign + approve
Complete required review

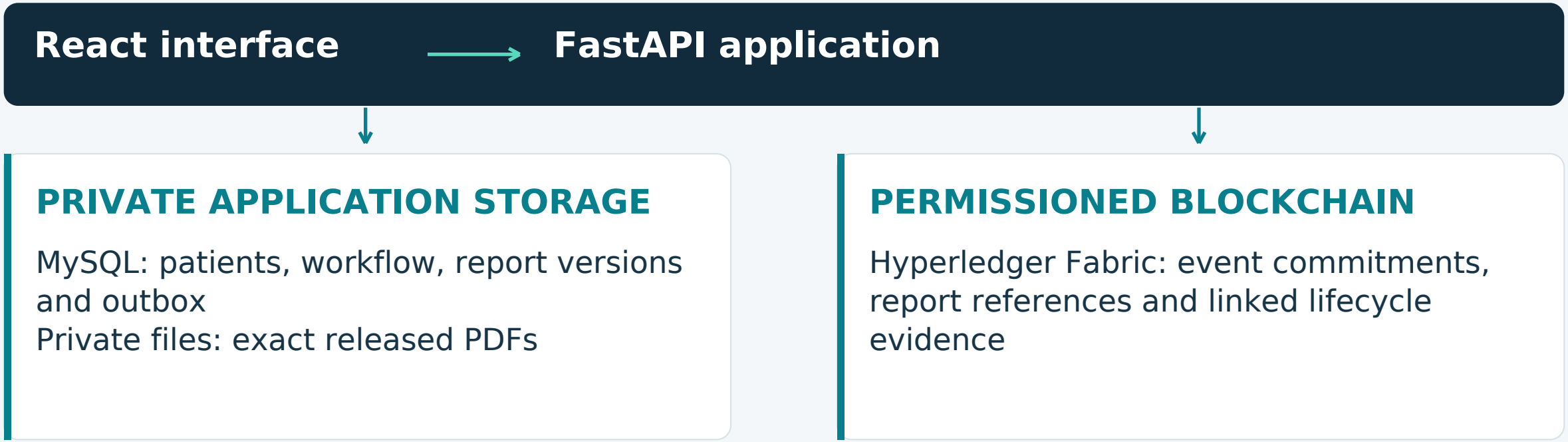
→

08
Release PDF
Issue report + capture event

Released reports are preserved; corrections follow a new version and approval cycle.

Paper: pp. 4, 18–20, 31–36.

Clinical records and blockchain evidence work together



Patient records and PDF files stay off-chain. Fabric stores verification anchors.

Paper: pp. 18–21; architecture diagram redrawn for this presentation.

Release captures the report and queues its evidence

01

Finalize PDF

QR + approved snapshot

→

02

Hash the PDF

SHA-256 of exact bytes

→

03

Build event

Identity + version + digest

→

04

Save release

Report state + outbox row

One database transaction

Report release and outbox capture commit together.
A background worker delivers the queued commitment.

A report can be RELEASED while its blockchain delivery is still PENDING.

Two hashes bind the PDF to its blockchain anchor

D = SHA-256(exact released PDF bytes)

D is the document fingerprint.

↓

Canonical event

Report UUID + version + D + event type + time + lifecycle links

↓

C = SHA-256(canonical event bytes)

Fabric anchor stores C as content_hash, with selected anchor metadata.

How an event becomes a committed Fabric transaction

01

Propose

Worker calls CreateAnchor

→

02

Endorse

Required organizations sign

→

03

Order

Orderer sequences transactions

→

04

Validate + commit

Peers validate and append

Chaincode checks

Allowed identity • valid fields • unique anchor ID

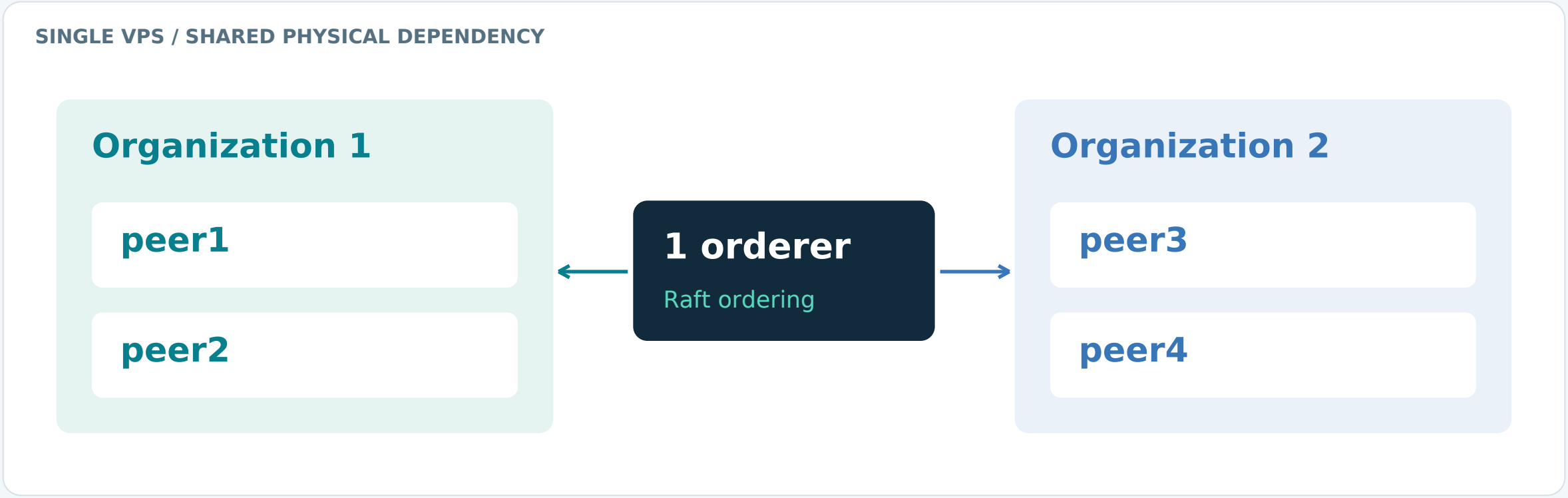
Confirmation receipt

Valid transaction • transaction ID • block reference

Transaction endorsement and later four-peer report verification are separate checks.

Paper: p. 20; gateway-adapter/src/adapter.js; network/config/configtx.single-vps.yaml.

Four peers, two organizations, one shared host



Logical distribution is demonstrated; independent-host resilience is not.

Paper: pp. 17, 20–21; blockchain/network/config/configtx.single-vps.yaml.

Verification joins document, lifecycle and ledger evidence

Scan QR / enter code → identify the registered report

Application checks

Registration
Current / revoked / superseded
Stored PDF + issuer evidence

Peer evidence

Read required report events
Check committed transaction
Compare four fresh observations

Submitted PDF

Optional file comparison
Hash the presented bytes
Compare with released digest

A recognized QR identifies the report; a matching PDF verifies the compared bytes.

Four-peer verification has explicit outcomes

4 / 4 matching	**VERIFIED** Complete, fresh report evidence from every peer.
3 / 4 + one offline	**PARTIAL** One peer cannot supply the required evidence.
Contradictory evidence	**CONFLICT** Matching peers do not override a contradiction.
Invalid / no evidence	**FAILED / UNAVAILABLE** The required proof cannot be accepted or obtained.

Network health and equal ledger heights do not establish report authenticity.

Paper: pp. 20–21, 25, 39–42; docs/FOUR_PEER_QR_VERIFICATION_COMPLETION.md.

A report’s history survives correction or withdrawal

Version 1 released

Official PDF v1
REPORT_RELEASED
Original event commitment

→

Version 2 released

New approval cycle
REPORT_RELEASED
New PDF + commitment

→

Version 1 replaced

REPORT_SUPERSEDED
Links to v1 release
References replacement

Withdrawal path REPORT_REVOKED links to the original release event.

Valid historical evidence does not mean the report is still current.

Paper: pp. 16, 20, 34; docs/PHASE_8B_TRANSACTIONAL_OUTBOX.md.

The queue separates clinical work from ledger delivery

01

Capture

Durable outbox event

→

02

Deliver

Submit through Gateway

→

03

Confirm

Check committed evidence

Temporary interruption

Retry with limits; reconcile an existing anchor before resubmission.

Conflict or exhausted retries

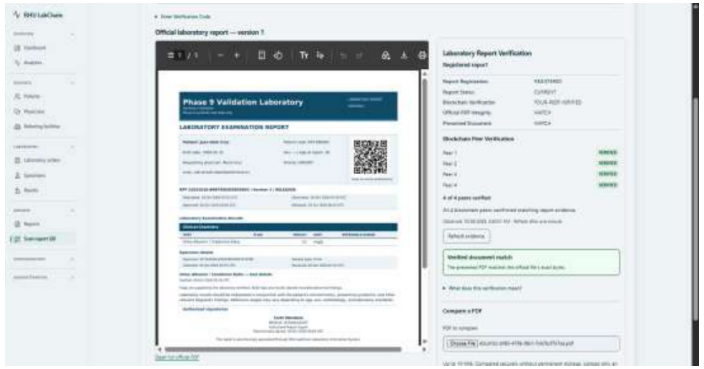
Keep the failure visible and require investigation.

PENDING is a delivery state. CONFIRMED is backed by a matching committed anchor.

Paper: pp. 18–20; docs/PHASE_8B_DELIVERY_WORKER.md.

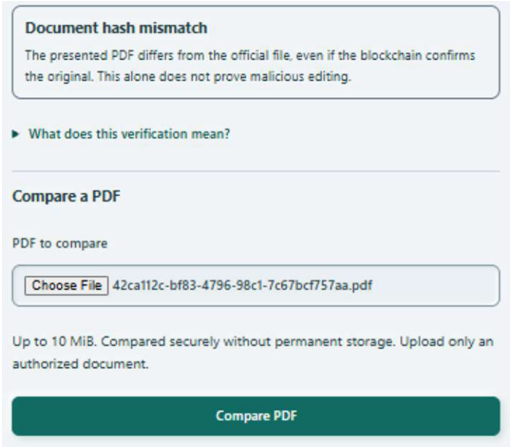
The paper shows successful and incomplete verification

Original PDF



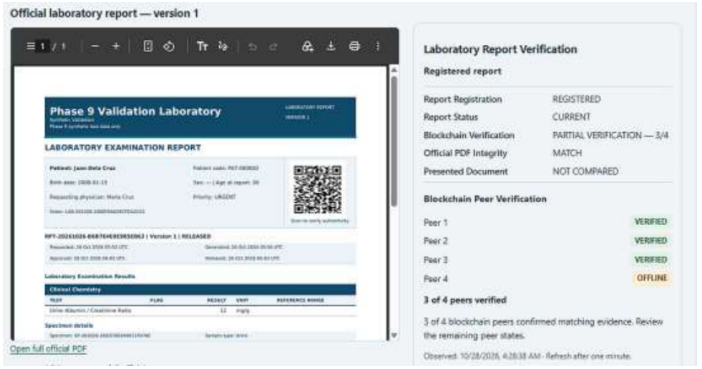
MATCH + four peers verified

Changed PDF



MISMATCH for submitted file

One peer offline



PARTIAL verification: 3/4

Observed interface states support the demonstrated cases; they are not benchmarks.

Paper: Fig. 4.9 (p. 37), Fig. 4.12 (p. 40), Fig. 4.14 (p. 41); synthetic report examples.

Survey feedback was favorable in both groups

4.64 / 5

Patients · n = 30

4.62 / 5

Staff · n = 11 (8 laboratory + 3 IT)

Research area	Patients	Staff
Security, privacy and integrity	**4.66**	**4.75**
Usability, accessibility and efficiency	**4.62**	**4.56**
User acceptance and trust	**4.64**	**4.53**

Descriptive perceptions on a 1–5 scale; groups analyzed separately.

Paper: Tables 4.3–4.4, pp. 42–44; corrected staff tally.

The prototype’s boundaries guide the next evaluation

Shared VPS and one orderer	→	Evaluate independent hosts and recovery.
Document integrity ≠ clinical accuracy	→	Review clinical workflow before real use.
Limited recorded technical scenarios	→	Record conflict, outage and access-denial tests.
Aggregate surveys; no timed comparison	→	Collect item-level data and task-time evidence.

Academic prototype using synthetic reports; clinical readiness remains future work.

Paper: pp. 20–25, 42, 47–50.

From a released report to evidence that can be checked

01 **CONTROL THE WORKFLOW**

Patient → order → specimen → result → release

02 **COMMIT THE EVIDENCE**

PDF fingerprint → canonical event → Fabric anchor

03 **VERIFY THE REPORT**

Lifecycle + document comparison + four-peer evidence

Questions & demonstration

Sources and presenter reference

Primary study

papers.pdf — 54 pages
System and blockchain model: pp. 18–21
Workflow and observed verification: pp. 31–42
Survey results: pp. 42–45 • Conclusions and limits: pp. 48–50

Implementation cross-checks

docs/PHASE_8B_TRANSACTIONAL_OUTBOX.md
docs/PHASE_8B_DELIVERY_WORKER.md
docs/FOUR_PEER_QR_VERIFICATION_COMPLETION.md
Fabric gateway adapter, anchor contract and single-VPS channel configuration

The paper’s findings and implementation explanations remain distinct.